CareFlow: Clinical Pathway Process Mining

Internship Project Report | Infotact Solutions | Ref. No. ITS/DA/1393 | Intern: Yash Rahangdale | Data Associate L1 Intern (Data Analytics) | 10 Oct 2026 - 10 Nov 2026

KPI	Value	KPI	Value
Patient cases	1,500	Clean events	13,583
Activities	15	DFG edges	28
Average length of stay	9.65 h	Long stay > 24 h	17 (1.13 %)
Worst bottleneck	Observation 226.6 min	Costliest dept	Operation Theatre
Total cost	INR 25,886,761	Avg cost / case	INR 17,258
Conformance fitness	1.0000	Long-stay AUC	0.998

1. Executive summary

CareFlow analyses a hospital event log with process-mining techniques and answers three operational questions: what is the REAL care pathway, where do time and money leak, and which patient will become a long stay. From 1,500 journeys and 13,583 clean events the project reconstructs 14 pathway variants on a 28-edge Directly-Follows Graph, locates the worst bottleneck (Observation, 226.6 min average wait), shows Operation Theatre consumes 36 % of cost (INR 9,381,243 of INR 25,886,761), and trains a long-stay classifier on leakage-free features (AUC 0.998, F1 0.800). Everything reproduces with one command.

2. Objectives

- O1 Discover the real pathway as a process model (DFG + variants).
- O2 Quantify bottlenecks, waiting time, service time, cost and resource load.
- O3 Measure conformance of actual cases against the expected pathway.
- O4 Predict long stay (LOS > 24 h) from leakage-free features.
- O5 Deliver an interactive dashboard and documentation to industry standards.

3. Data

Event log: one row = one event. Mandatory case_id (patient journey), activity, timestamp; optional resource, department, wait_minutes, service_minutes, cost_inr. Case attributes: age, gender, triage_level, admission_type, pathway, los_hours, total_cost_inr, n_events.

Property	Value
Raw rows / clean rows	13,664 / 13,583 (81 duplicates removed)
Cases / activities	1,500 / 15
Period	Mar 2026 - Aug 2026
Engineered columns	26 (event_log_clean.csv)

Provenance: the reference log is realistic but synthetic (src/generate_data.py, seed 42) because patient-confidential data cannot be used. For real data the project documents the public 4TU.ResearchData 'Sepsis Cases - Event Log' (Mannhardt; ~1,050 cases / 15,214 events / 16 activities) with the exact XES-to-CSV column mapping, so the same pipeline runs unchanged.

4. Pipeline architecture

```
data/raw/event_log.csv
  |  clean_data.py      8 cleaning rules + audit trail
  v
data/processed/event_log_clean.csv   (26 features)
  |- discover_models.py  DFG (28 edges), 14 variants
  |- analyze_insights.py KPIs, bottlenecks, 14 PNG charts
  |- conformance.py      token-based replay fitness
  |- ml_model.py         long-stay classifier + evaluation
  |- connect_db.py       MySQL / PostgreSQL / SQLite star schema
  `- dashboard/app.py    Streamlit dashboard
run_all.py  ->  all six stages, ~25 seconds
```

5. Data cleaning rules

#	Rule	Justification	Effect
1	Trim/lower names, cast types	consistent schema	8 columns
2	Drop rows with no case_id/activity/timestamp	unmineable	0 rows
3	Remove exact duplicate events	double-scanned events inflate counts	81 rows
4	Blank resource -> UNKNOWN, dept -> Unassigned	never delete an event for a label	203 filled
5	Negative wait/cost -> 0; service <= 0 -> median	impossible / robust	outliers tamed
6	Derive event_index, gap_minutes, is_rework	core mining features	3 columns
7	Drop cases < 3 or > 200 events	too short / corrupt	0 cases

6. Results

6.1 Top directly-follows edges

From - To	Count	From - To	Count
Registration -> Triage	1,100	Emergency Assessment -> Lab Test	579
Discharge Planning -> Discharge	979	Radiology -> Diagnosis Confirmed	579
Specialist Consultation -> Lab Test	921	Triage -> Emergency Assessment	579
Physiotherapy -> Discharge Planning	775	Lab Test -> Diagnosis Confirmed	521
Diagnosis Confirmed -> Medication	725	Medication Prescribed -> Discharge	521

6.2 Department cost concentration

Department	Events	Cost (INR)	Department	Events	Cost (INR)
Operation Theatre	386	9,381,243	ICU	228	1,528,075
Ward	1,951	4,903,160	Rehabilitation	862	1,254,142
Laboratory	1,690	2,526,299	Front Desk	3,018	721,690
Outpatient	2,034	1,783,783	Pharmacy	1,128	565,041
Radiology	680	1,675,840	Emergency	1,687	1,649,701

6.3 Model evaluation (long stay > 24 h)

Two feature sets were compared deliberately. The 'full' set contains post-outcome aggregates (they exist only after the stay ends) and therefore scores a perfect 1.000 - a textbook leakage artefact. The leakage-free set (age, gender, triage level, admission type, pathway, event count) is the shipped production model.

Feature set	Model	Accuracy	F1	ROC-AUC	CV F1
full (leaky)	logistic regression	1.000	1.000	1.000	0.727 +/- 0.167
full (leaky)	random forest	1.000	1.000	1.000	1.000 +/- 0.000
full (leaky)	gradient boosting	1.000	1.000	1.000	1.000 +/- 0.000
leakage-free	logistic regression	0.993	0.500	0.987	0.080 +/- 0.160
leakage-free	random forest	0.990	0.000	0.996	0.000 +/- 0.000
leakage-free (shipped)	gradient boosting	0.997	0.800	0.998	0.180 +/- 0.223

Honest reading: long stays are only 1.13 % of cases (17 of 1,500), so accuracy misleads and cross-validated F1 swings widely. The model ranks risk well (AUC 0.998) but is not yet a clinically reliable decision rule. Next step: more long-stay cases or cost-sensitive learning / SMOTE, and validation on a real log where AUC 0.75-0.90 is expected.

7. Business recommendations

- Fix Observation first - 226.6 min average wait is the largest recoverable delay.
- Shift elective activity away from Friday 14:00 (peak load).
- Audit Operation Theatre utilisation - 36 % of total cost sits in one department.
- Standardise the most repeated step to remove rework at source.
- Fast-track Emergency Assessment -> Lab Test (579 cases) with a pre-approved lab panel.
- Use the risk score at triage to prioritise probable long stays for case management.

8. Limitations and next steps

- Reference log is synthetic; real-log fitness is expected at 0.7-0.95, not 1.0.
- Cost is activity-level, not invoice-level; no payer dimension.
- Extreme class imbalance (1.13 %) limits classifier reliability.
- Next: real HIS log, pm4py Inductive Miner + Petri net, dashboard deployment, process drift monitoring.

9. Reproducibility

```
pip install -r requirements.txt
python run_all.py                 # 6 stages, ~25 s
python src/connect_db.py --engine sqlite
streamlit run dashboard/app.py
```

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